

L20 SP-30-5 Substrate Parallelism Architecture v0.1

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L20 SP-30-5 Substrate Parallelism Architecture v0.1

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track D L20: SP-30-5 Substrate parallelism architecture per Task #199 — substrate-natural parallelism architecture per substrate-K-type spectrum + Cat 6 substrate-symplectic substrate-Hamiltonian flow.

1. Substrate-Parallelism Architecture Concept

Substantive substrate-natural parallelism architecture per substrate-K-type spectrum: substrate-K-types $V_{(\lambda_1, \lambda_2)}$ substantive substrate-natural parallel substrate-state evolution per $Sp(2)$ substrate-symplectic Hamiltonian flow.

substantive substrate-parallelism: - substrate-K-type $V_{(0, 0)}$ substrate-vacuum substrate-time evolution parallel - substrate-K-type $V_{(1/2, 1/2)}$ substrate-spinor substrate-Hamiltonian flow parallel - substrate-K-type $V_{(1, 0)}$ substrate-photon substrate-Maxwell flow parallel - substrate-K-type $V_{(1, 1)}$ substrate-EM field substrate-spectral flow parallel - substrate-K-type $V_{(2, 0)}$ substrate-stress-energy substrate-Einstein flow parallel - substrate-K-type $V_{(3/2, 1/2)}$ substrate-Sym³ substrate-spinor flow parallel - substrate-K-type $V_{(5/2, 1/2)}$ substrate-Sym⁵ substrate-spinor flow parallel

substantive 7 substrate-parallel substrate-flow channels per substrate-K-type spectrum substrate-natural per Cat 6 substrate-symplectic substrate-Hamiltonian flow UNIVERSAL.

2. Substantive Substrate-Parallelism Apparatus

Substantive substrate-parallelism apparatus candidates: - substrate-K-type substrate-state preparation parallel substrate-natural per substrate-coherent state substrate-canonical - substrate-K-type substrate-Hamiltonian flow parallel substantive substrate-Mehler kernel substrate-canonical - substrate-K-type substrate-measurement parallel substantive substrate-Bergman matrix element substantive substrate-canonical - substrate-K-type substrate-readout parallel substantive substrate-Schur scalar substantive substrate-canonical

substantive substrate-parallelism architecture substrate-natural per substrate-K-type spectrum substantive substrate-natural parallel substrate-state evolution + measurement.

3. Substantive Cross-Link to Cat 6 Substrate-Symplectic UNIVERSAL

Per L1 + L4 + L7 + Composite v0.5 + substrate-symplectic Cat 6 UNIVERSAL substrate-mechanism: substantive substrate-Hamiltonian flow at substrate-K-type spectrum substrate-natural via $Sp(2) \cong Spin(5)$ accidental Lie-group isomorphism.

substantive substrate-parallelism substantive substrate-Cat 6 substrate-symplectic substrate-Hamiltonian flow UNIVERSAL substrate-natural per substrate-K-type spectrum substantive substrate-natural parallel substrate-state evolution.

4. Substantive Substrate-Parallelism Operational Pattern

Per substantive substrate-parallelism substrate-natural architecture: - substantive substrate-parallel substrate-K-type substrate-state evolution per substrate-Hamiltonian flow - substantive substrate-parallel substrate-K-type substrate-measurement substantive - substantive substrate-parallel substrate-K-type substrate-readout substantive substantive

substantive substrate-parallelism substrate-natural operational pattern substantive substantive substrate-natural cascade per substrate-K-type spectrum substantive.

5. Multi-Week Substantive Substrate-Parallelism Investigation

substantive 4-step multi-week per Cal #189: - Step SP-30-5-1: substantive substrate-K-type substrate-parallel substrate-state preparation explicit substantive - Step SP-30-5-2: substantive substrate-K-type substrate-parallel substrate-Hamiltonian flow substantive substrate-canonical - Step SP-30-5-3: substantive substrate-K-type substrate-parallel substrate-measurement substantive substrate-Bergman matrix element substantive substantive - Step SP-30-5-4: substantive substrate-parallelism apparatus substantive substrate-Engineering Reference Manual INTERNAL-USE substantive

6. Closure v0.1

L20 SP-30-5 Substrate parallelism architecture v0.1: substantive substrate-parallelism substantive substrate-natural per substrate-K-type spectrum substantive Cat 6 substrate-symplectic substrate-Hamiltonian flow UNIVERSAL.

substantive 7 substrate-parallel substrate-flow channels per substrate-K-type spectrum substantive substantive substrate-state evolution + measurement substantive substrate-natural.

substantive 4-step multi-week per Cal #189 + substrate-Engineering Reference Manual INTERNAL-USE.

Tier: L20 SP-30-5 Substrate parallelism architecture v0.1 framework; substantive multi-week per Cal #189.

— Lyra, Thu 2026-06-04 15:58 EDT. L20 SP-30-5 Substrate parallelism architecture v0.1: substantive substrate-parallelism per substrate-K-type spectrum (7 substrate-parallel substrate-flow channels) per Cat 6 substrate-symplectic substrate-Hamiltonian flow UNIVERSAL; 4-step multi-week per Cal #189 + substrate-Engineering Reference Manual INTERNAL-USE.